1

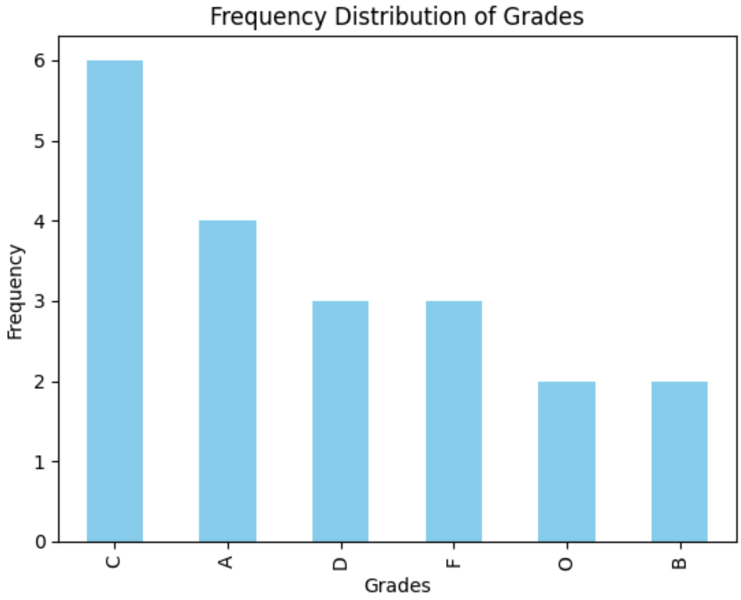
```
import pandas as pd

# Load dataset
df = pd.read_csv("employee.csv")

# Number of rows and columns
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])

# Data types
print("\nData Types:")
print(df.dtypes)

# Missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Statistical summary
print("\nStatistical Summary:")
print(df.describe(include='all'))
```

```
Number of Rows: 20
Number of Columns: 5

Data Types:
EmpID           int64
Name           object
Age             int64
Department     object
Salary        float64
dtype: object

Missing Values:
EmpID         0
Name          0
Age           0
Department    0
Salary        1
dtype: int64

Statistical Summary:
            EmpID  Name        Age Department         Salary
count    20.00000    20  20.000000         20      19.000000
unique        NaN    20        NaN          4            NaN
top           NaN  Emp1        NaN         IT            NaN
freq          NaN     1        NaN          6            NaN
mean    110.50000   NaN  32.050000        NaN   52315.789474
std       5.91608   NaN   8.678315        NaN   25599.593381
min     101.00000   NaN  22.000000        NaN   32000.000000
25%     105.75000   NaN  24.750000        NaN   35000.000000
50%     110.50000   NaN  29.000000        NaN   45000.000000
75%     115.25000   NaN  40.500000        NaN   55000.000000
max     120.00000   NaN  45.000000        NaN  120000.000000
```

2

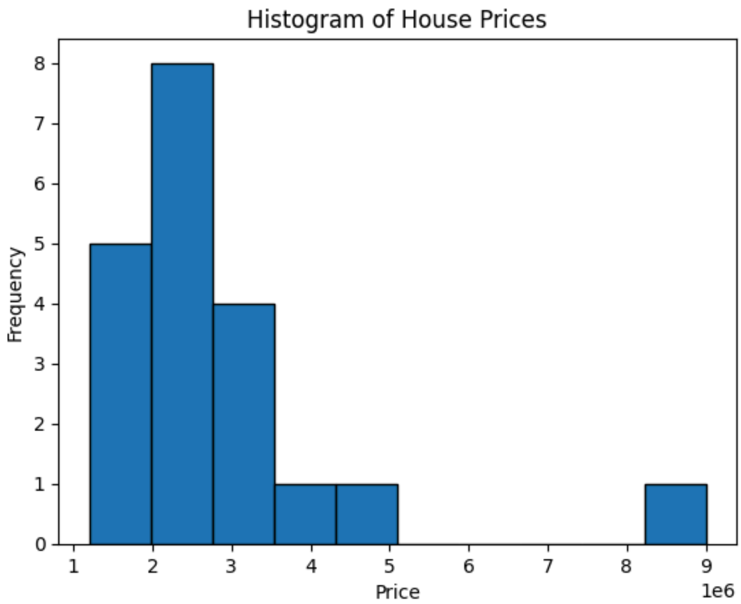
```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("students.csv")

# Frequency distribution
grade_freq = df["Grade"].value_counts()

print(grade_freq)

# Bar chart
grade_freq.plot(kind="bar", color="skyblue")
plt.title("Frequency Distribution of Grades")
plt.xlabel("Grades")
plt.ylabel("Frequency")
plt.show()
```

```
Grade
C    6
A    4
D    3
F    3
O    2
B    2
Name: count, dtype: int64
```



3

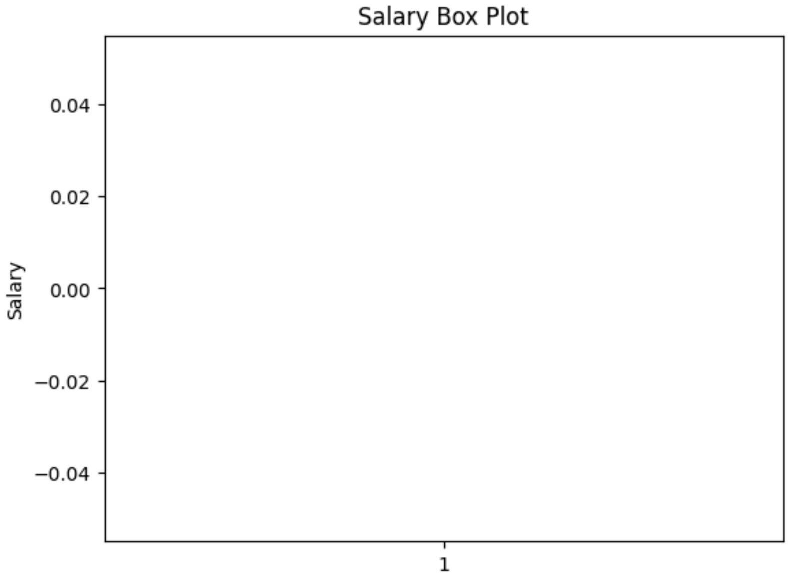
```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("house.csv")

# Histogram
plt.hist(df["Price"], bins=10, edgecolor="black")
plt.title("Histogram of House Prices")
plt.xlabel("Price")
plt.ylabel("Frequency")
plt.show()

# Skewness
print("Skewness:", df["Price"].skew())

if abs(df["Price"].skew()) < 0.5:
    print("Approximately Normally Distributed")
else:
    print("Skewed Distribution")
```




```
Skewness: 2.8498062050339947
Skewed Distribution
```

4

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("employee.csv")

# Box plot
plt.boxplot(df["Salary"])
plt.title("Salary Box Plot")
plt.ylabel("Salary")
plt.show()

# Outliers
Q1 = df["Salary"].quantile(0.25)
Q3 = df["Salary"].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df["Salary"] < lower) | (df["Salary"] > upper)]

print("Outliers:")
print(outliers)
```




```
Outliers:
   EmpID  Name  Age Department    Salary
4    105  Emp5   30         IT  120000.0
8    109  Emp9   45         IT  120000.0
```

5

```
import pandas as pd

# Load dataset
df = pd.read_csv("students.csv")

Q1 = df["Marks"].quantile(0.25)
Q3 = df["Marks"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df["Marks"] < lower) | (df["Marks"] > upper)]

print("Outliers:")
print(outliers)
```

```
Outliers:
Empty DataFrame
Columns: [StudentID, Name, Marks, Grade]
Index: []
```

6

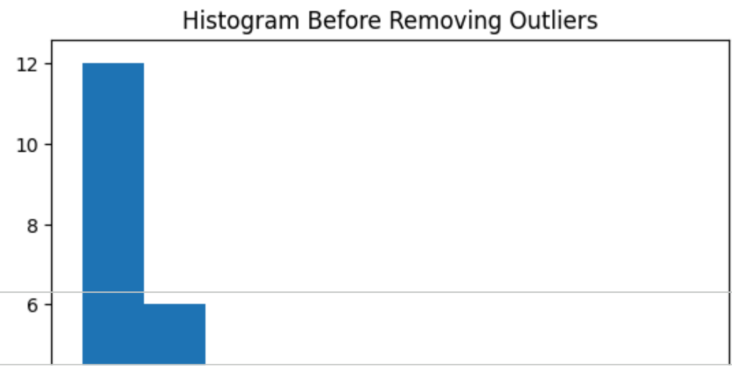
```
import pandas as pd

# Load dataset
df = pd.read_csv("sales.csv")

Q1 = df["Sales"].quantile(0.25)
Q3 = df["Sales"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

cleaned_df = df[(df["Sales"] >= lower) & (df["Sales"] <= upper)]

print("Cleaned Dataset:")
print(cleaned_df)
```

```
Cleaned Dataset:
   Product  Sales
0       P1    210
1       P2    220
2       P3    215
```

```
3       P4    230
4       P5    240
5       P6    250
6       P7    245
7       P8    235
8       P9    225
9      P10    260
10     P11    255
11     P12    248
12     P13    238
13     P14    242
16     P17    230
17     P18    228
18     P19    235
19     P20    240
```

7

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("sales.csv")

Q1 = df["Sales"].quantile(0.25)
Q3 = df["Sales"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

cleaned = df[(df["Sales"] >= lower) & (df["Sales"] <= upper)]

# Histogram Before
plt.hist(df["Sales"], bins=10)
plt.title("Histogram Before Removing Outliers")
plt.show()

# Histogram After
plt.hist(cleaned["Sales"], bins=10)
plt.title("Histogram After Removing Outliers")
plt.show()

# Box Plot Before
plt.boxplot(df["Sales"])
plt.title("Box Plot Before Removing Outliers")
plt.show()

# Box Plot After
plt.boxplot(cleaned["Sales"])
plt.title("Box Plot After Removing Outliers")
plt.show()
```



8

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris

# Load Iris dataset
iris = load_iris(as_frame=True)

df = iris.frame

# Dataset Information
print("Dataset Info:")
print(df.info())

# Missing Values
print("\nMissing Values:")
print(df.isnull().sum())

# Descriptive Statistics
print("\nDescriptive Statistics:")
print(df.describe())

# Histograms
df.hist(figsize=(8,6))
plt.show()

# Box Plots
df.plot(kind="box", figsize=(8,6))
plt.show()

# Detect and Remove Outliers
numeric_cols = df.select_dtypes(include='number').columns

for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    df = df[(df[col] >= lower) & (df[col] <= upper)]

print("\nCleaned Dataset:")
print(df)

# Save Cleaned Dataset
df.to_csv("iris_cleaned.csv", index=False)

print("Cleaned dataset saved as iris_cleaned.csv")
```